55,000 passenger vehicles, and a million two-wheelers. The FAME II scheme has now been extended till 31st March, 2024 and only about 10% of the subsidy has been spent so far, which means there is lot of scope left for prospective buyers to get EVs at subsidized costs right at the dealership. States can further add-on their own incentivization while also wooing EV players to set up shop in their states for manufacturing. However, delays in re-certification process of existing EVs to FAME II and a plan to tax non-electric vehicles heavier to drive EV sales has drawn a lot of flak.

The Government of India also announced a new Production Linked Incentive for Advanced Chemistry Cell Battery Storage (PLI-ACC) in 2021 with an outlay of 18,000 crore. The PLI-ACC scheme aims to attract local and global investors to invest in ACC facilities in the country. The country's policy thinktank, Niti Ayog, expects to achieve 50 Giga Whr capacity in the next five years.

The Finance Ministry, in its latest budget presentation, proposed lowering the customs duties on nickel ore and concentrates from 5% to 0%, nickel oxides from 10% to 0%, ferro nickel from 15% to 2.5% with the hope that this will reduce lithium-ion battery production costs. The Ministry also proposed lowering duties on motor parts from 10% to 7.5%.

PERCEPTIONAL CHALLENGES

The rate of EV adoption is set to increase in the post-COVID era, but it does take conscious effort to change the perception of a population, especially one that is as diverse as India's. While there has been considerable buzz about the economic and environmental advantages of going electric despite the higher upfront costs, there has also been some negative press and various social media outpourings about the quality of EVs being currently made.

The major perceptional barriers come via concerns about availability of skilled service and safety. ICE vehicles of manufacturers such as Maruti-Suzuki, Hero Honda, TVS, and Royal Enfield among others are popular not only for their flagship and best-selling models but also for ease of service. Even a run-of-the-mill local mechanic shop garage has the required know-how to service these vehicles given their ubiquitous nature, which can prove to be a blessing during unexpected breakdowns on the road. EVs are still a nascent market, so wide availability of spare parts and ease of finding any roadside garage is still a pipedream at the moment.

Then there's also the perception about real-world performance and mileage. ICE vehicles have now matured-enough to offer excellent mileage often nearing the manufacturer's claims. When it comes to EVs, the range is determined based on benchmark data from testing agencies, under controlled test conditions that may not mimic real-world usage. Also, range anxiety subconsciously motivates the driver to operate the vehicle at lower speeds than what it is capable of. A good deal of achieving this range also factors regenerative braking.

Quality of parts that go into an EV is another major concern. Manufacturers seems to be focused more on adding new tech such as touchscreen panels, GPS, smartphone connectivity, and delicate sensors but seem to be losing sight of the actual parts that make a vehicle tick. Most of these parts are custom-made, which means that you will have to inevitably run to the authorized dealership and pay a premium to get them replaced.

In this day and age, social media plays a huge role in molding people's perceptions. Recently, the web was abuzz with reports of electric vehicles catching fire for no apparent reason and auto-reversing at high speeds due to a software glitch. In the wake of such increasing number of incidents, the government had to step in and issue a stern warning with threat of heavy penalties and recalls. Companies such as Ola, Pure EV, and Okinawa have had to recall nearly 6,700 of their electric scooters following the government's diktat. These accidents and ensuing loss and personal injury have dissuaded many a prospective buyer from opting for EVs just yet.

Any large-scale transition does have its pitfalls. While teething problems do exist, the benefits of going electric do have a significant impact in the long run to alleviate India's excessive reliance on imported fossil fuels and providing cheap and smarter ways of commute. Many of the perceptional issues are likely to get annulled in the due course of time once economies of scale is attained.

If the government continues to incentivize EV adoption and companies can prove that their EV quality is on par or better than the ICE equivalents, India offers a large untapped market just waiting to be explored. d